

# DVF QA metrics — phase 70

| Metric                       | Value      |
|------------------------------|------------|
| qa_status                    | FAIL       |
| jac_min                      | -0.427098  |
| jac_p01                      | 0.65757    |
| jac_p05                      | 0.798885   |
| jac_median                   | 0.997328   |
| jac_mean                     | 0.998627   |
| jac_p95                      | 1.20565    |
| jac_p99                      | 1.34974    |
| jac_max                      | 3.19295    |
| fraction_jac_le_0            | 6.4e-05    |
| fraction_jac_lt_0_2          | 0.00040448 |
| fraction_jac_gt_5            | 0          |
| folding_component_count      | 602        |
| largest_folding_component_ml | 0.008      |
| dvf_magnitude_p95_mm         | 4.57938    |
| dvf_magnitude_max_mm         | 10.0668    |
| gradient_norm_mean           | 0.2038     |
| bending_energy_mean          | 0.0383204  |
| warped_vs_fixed_mse          | 6755.99    |
| warped_vs_fixed_mae          | 26.513     |
| warped_vs_fixed_ncc          | 0.985718   |
| our_warp_vs_precomputed_mse  | 6755.99    |
| our_warp_vs_precomputed_mae  | 26.513     |
| our_warp_vs_precomputed_ncc  | 0.985718   |

CT comparison (moving / fixed / warped / |diff|)

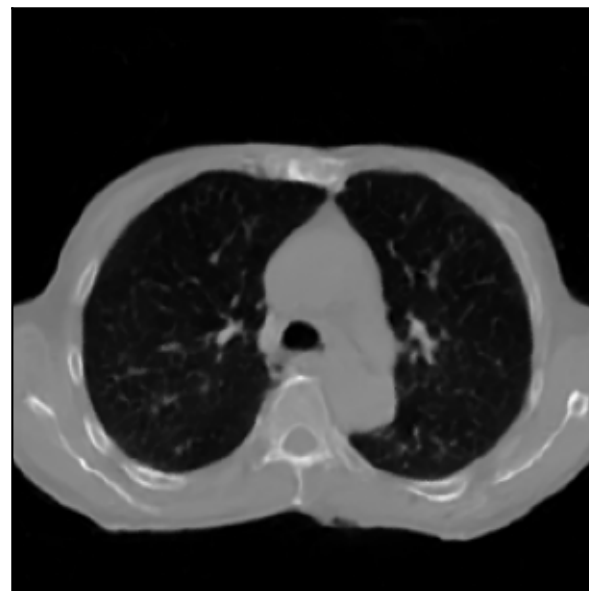
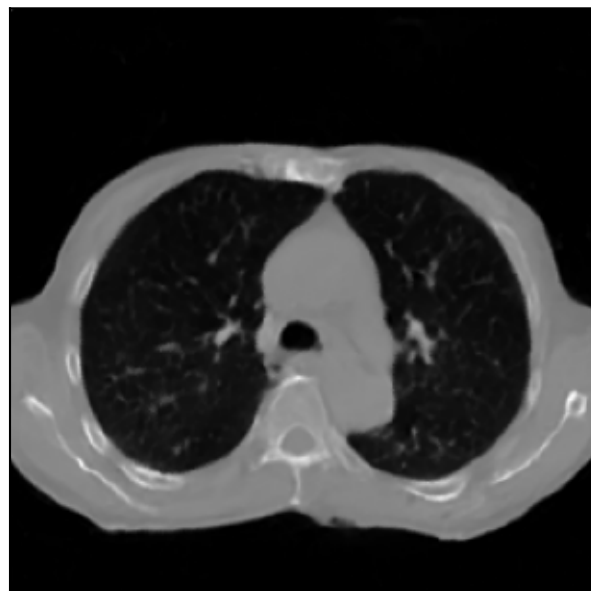
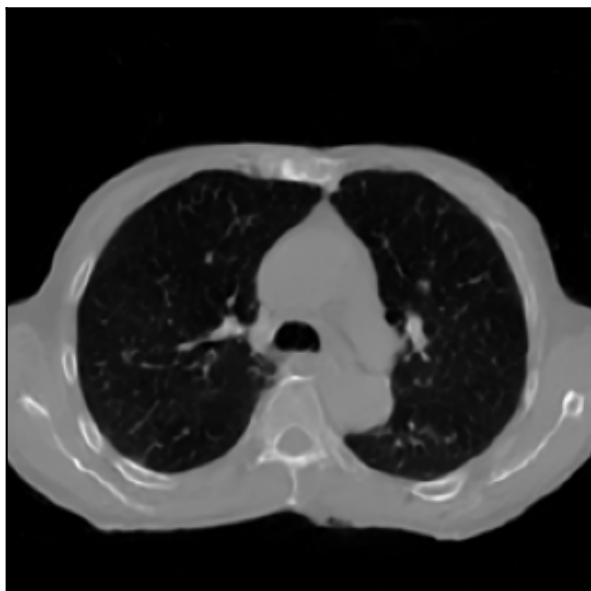
Moving (phase 50)

Fixed (phase 70)

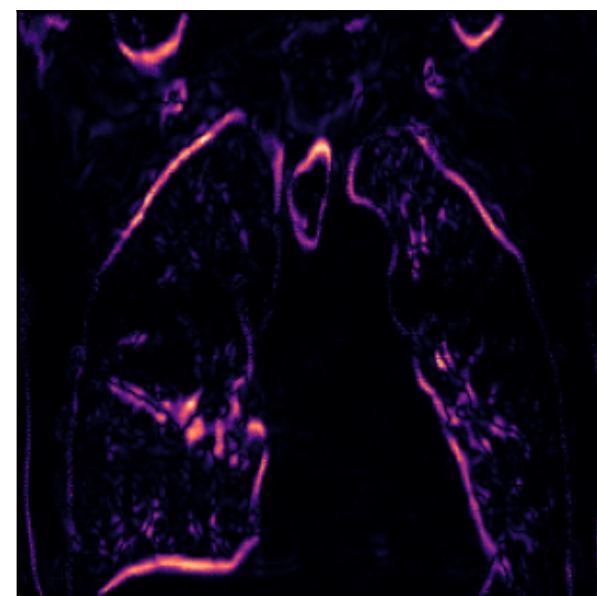
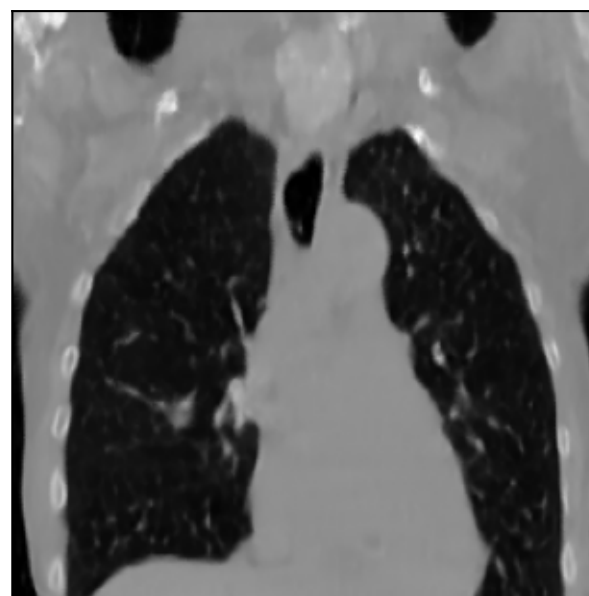
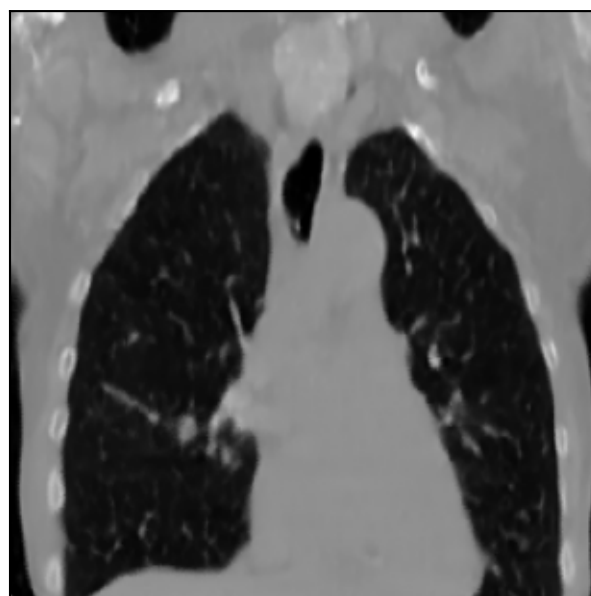
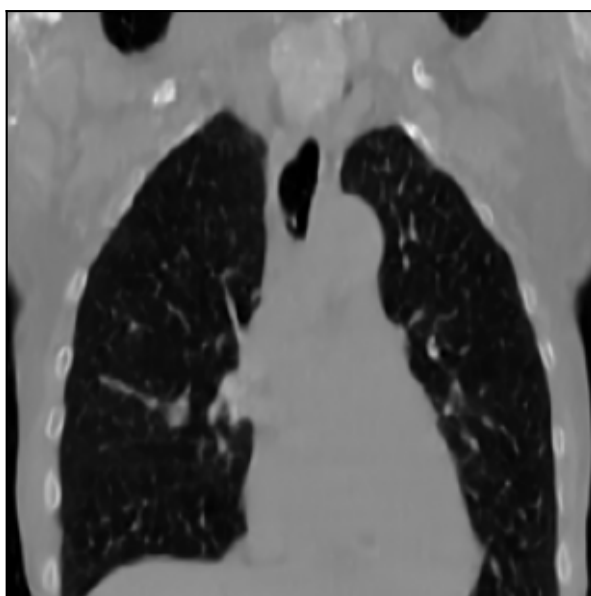
Warped (50→70)

|Fixed - Warped|

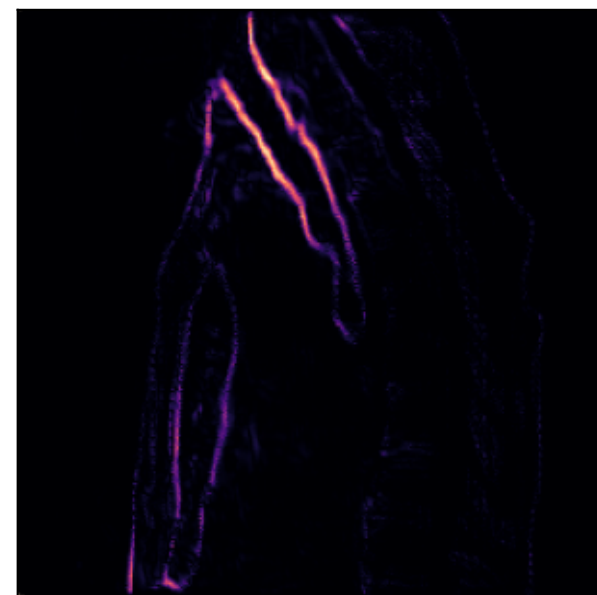
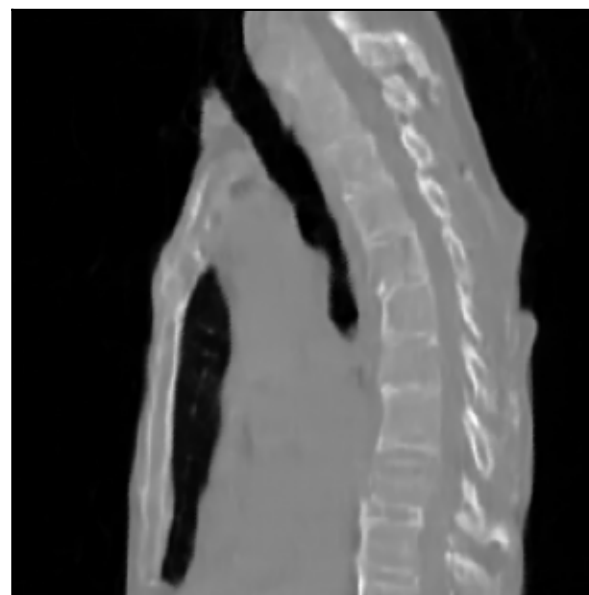
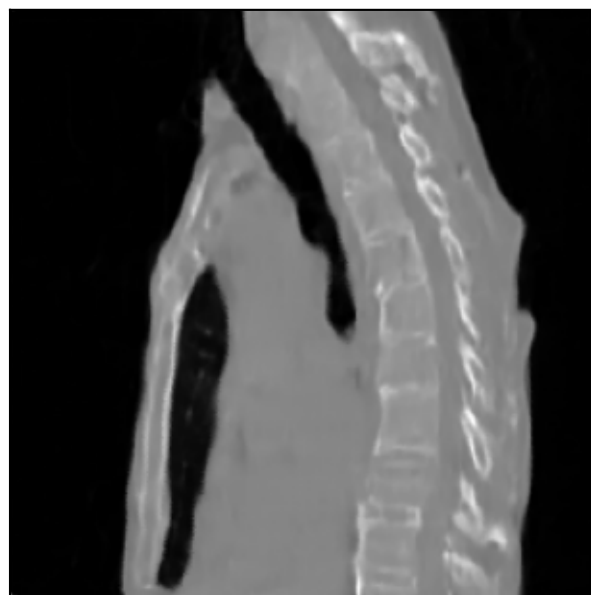
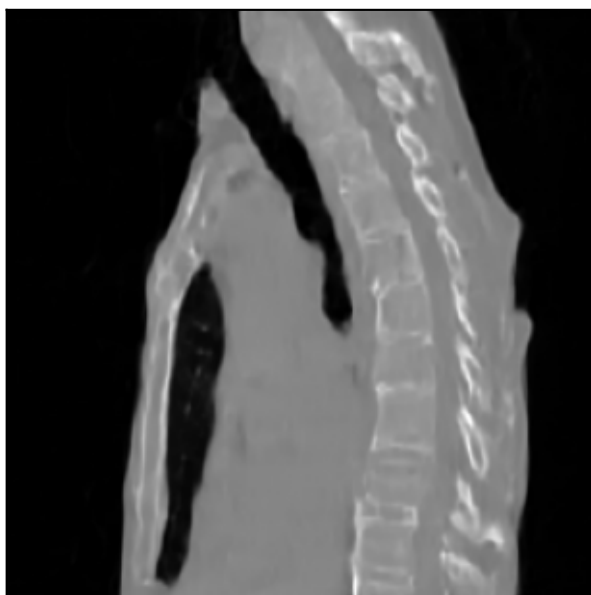
Axial mid



Coronal mid

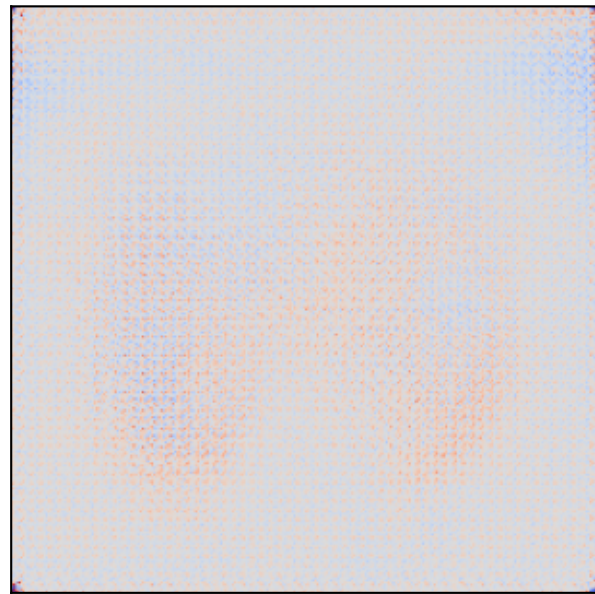


Sagittal mid

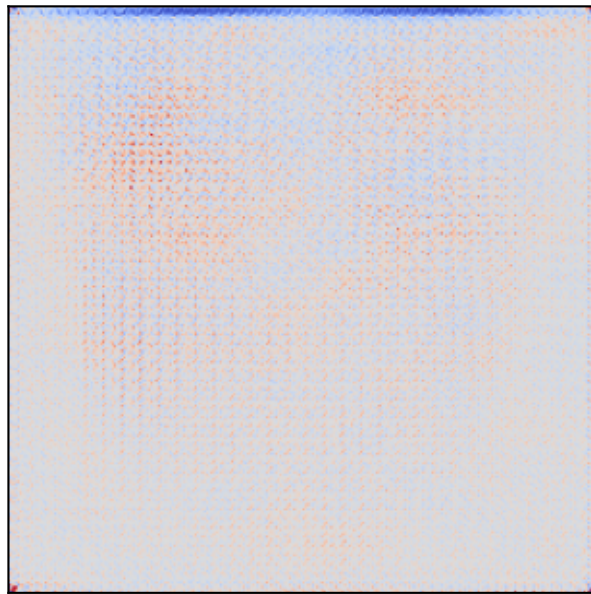


Jacobian determinant + histogram

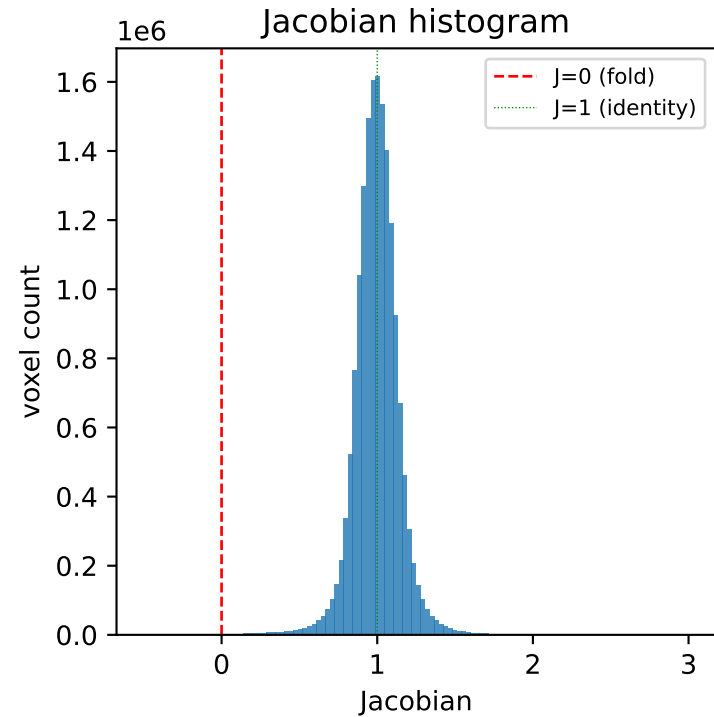
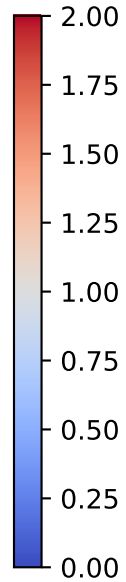
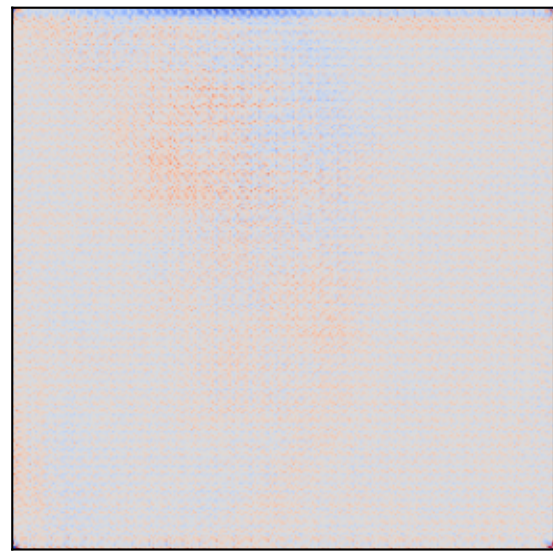
Jacobian Axial mid



Jacobian Coronal mid

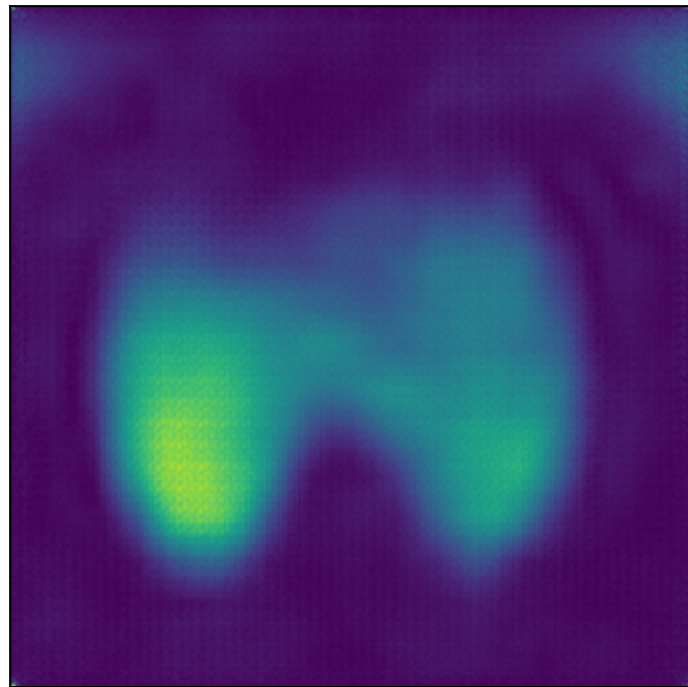


Jacobian Sagittal mid

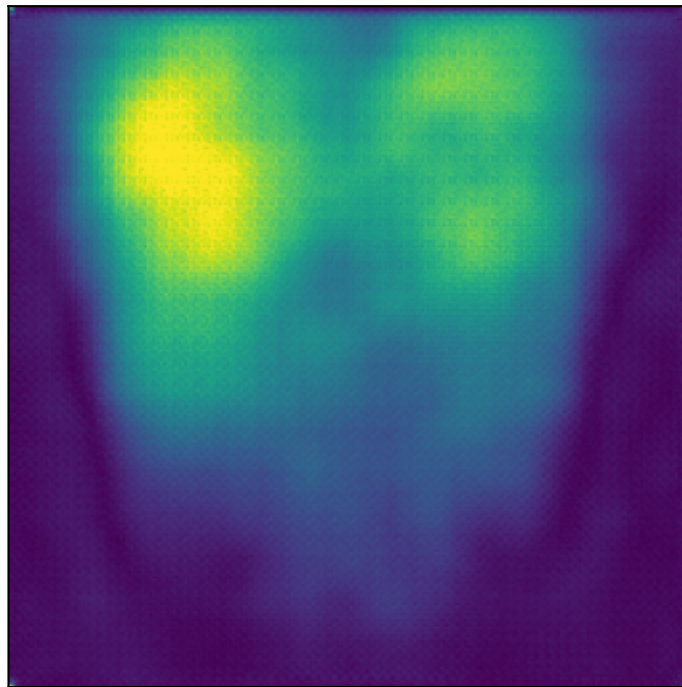




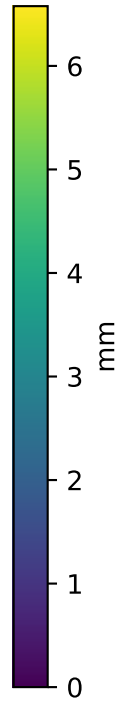
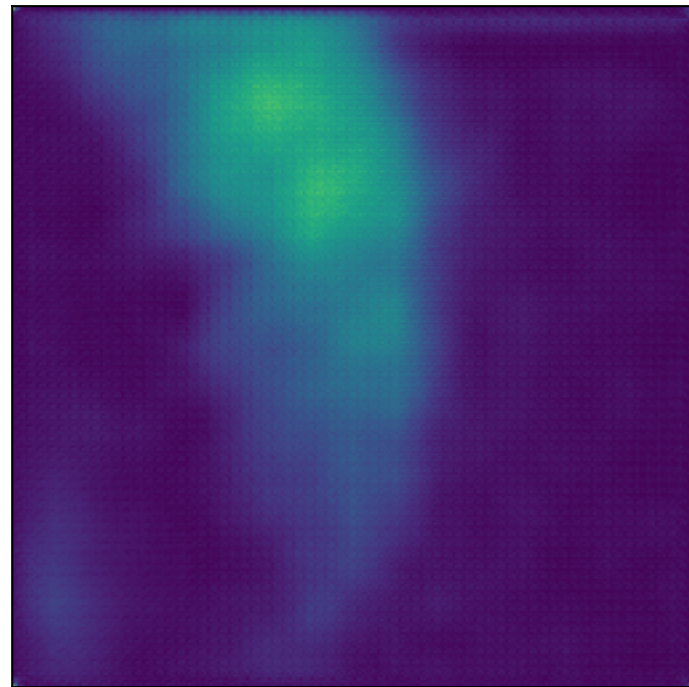
$|u|$  Axial mid



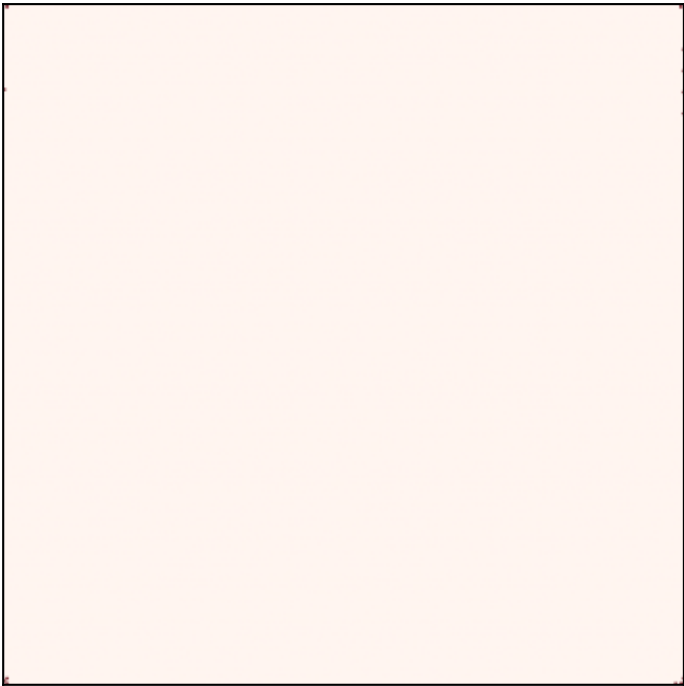
DVF magnitude  
 $|u|$  Coronal mid



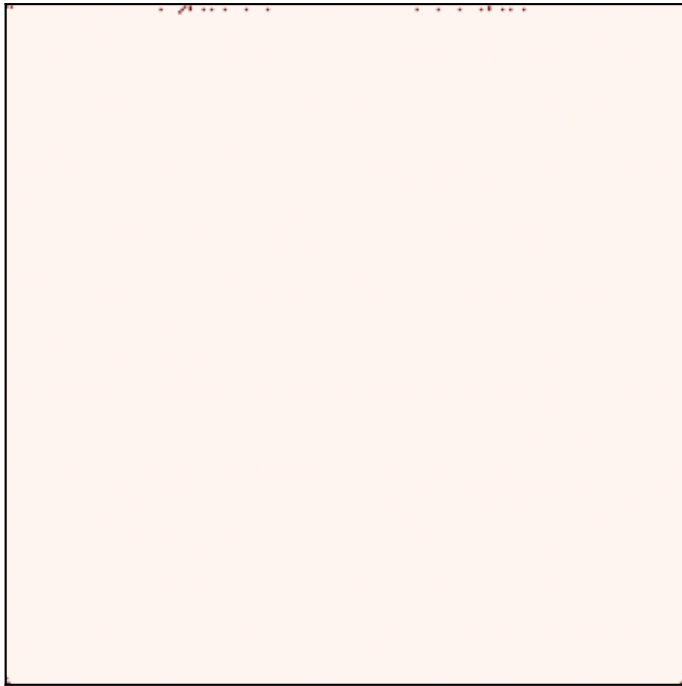
$|u|$  Sagittal mid



Folding mask Axial mid



Folding voxels ( $J \leq 0$ )  
Folding mask Coronal mid



Folding mask Sagittal mid

